PROJECT PROPOSAL: PROJECT AXIS BIO-REFINERY

Submitting Entities: Odynex Axis Logistics LLC, SOWEGA Asian Pacific Chamber of Commerce and Foundation, & Odynex Institute
Proposed Location: Fort Gaines / Clay County, Southwest Georgia (SOWEGA)
Operational Footprint: 100% Indoor, Zero-Environmental-Discharge, Bio-Based Manufacturing

I. Executive Summary

Project Axis is a highly sustainable, closed-loop urban agriculture and biochemical processing initiative spearheaded by Odynex Axis Logistics LLC, supported by the community and global networks of the SOWEGA Asian Pacific Chamber of Commerce and Foundation, and scientifically driven by the Odynex Institute. The facility utilizes controlled, indoor-bred insect biomass to extract Chitin/Chitosan (a high-value biopolymer for medical, water purification, and industrial use) and manufacture Probiotic Bone Meal & Organic Soil Amendments for the regional agricultural and poultry sectors. By replacing harsh synthetic chemicals with natural microbial fermentation and underutilized local agricultural waste streams, this facility operates with a zero-environmental-discharge footprint in Fort Gaines, Georgia.

II. Organizational Structure & Roles

- **Odynex Axis Logistics LLC:** Serves as the primary operational engine managing property acquisition, facility operations, regional waste-stream sourcing, and B2B product distribution across the Southeastern United States.
- **SOWEGA Asian Pacific Chamber of Commerce and Foundation:** Drives community engagement, coordinates with regional economic development authorities, and establishes global export pipelines to tap into the high international demand for sustainable biopolymers and organic agricultural inputs.
- **Odynex Institute:** Functions as the dedicated research, development, and intellectual property hub, focusing on the optimization of green, chemical-free enzymatic extraction methods, biorefinery scaling, and product safety testing.

III. Core Operational Technology & Biosecurity

- **Controlled Biomass Cultivation:** Closed-environment, tropical insect arrays (*Blaptica dubia* or *Blaberus discoidalis*) housed within a fully indoor, climate-controlled vertical facility. Operations utilize multi-layered physical containment, slick-walled barrier arrays, and fine-mesh ventilation screens. These non-native species cannot survive local winter temperatures, entirely eliminating environmental escape hazards.
- **Green Bio-Extraction Loop:** The Odynex Institute's proprietary biological deproteinization process utilizes natural *Lactobacillus* fermentation powered by underutilized local agricultural carbon sources (e.g., fruit pulp, molasses). Demineralization utilizes mild organic acids cultivated from regional crop residues (e.g.,

peanut hull substrates), completely bypassing the need for hazardous synthetic acids or bases.

IV. Infrastructure & Environmental Footprint

- **Property Strategy:** Adaptive re-use of an existing rural commercial warehouse, light industrial space, or agricultural facility within Clay County, bringing vacant real estate back onto active tax rolls.
- **Zero-Discharge Closed Loop:** The extraction process utilizes an indoor water recycling, distillation, and ultrafiltration loop managed by Odynex Axis Logistics LLC. Zero industrial wastewater is discharged into the municipal sewer or local water tables, eliminating the requirement for complex National Pollutant Discharge Elimination System (NPDES) permits.
- **Air Quality & Odor Control:** Industrial-grade HEPA air filtration and dust-collection systems will continuously capture airborne particulates (chitin dust and frass), protecting worker respiratory health in full compliance with OSHA standards while completely neutralizing exterior odors.

V. Regional Synergy & Inter-County Partnerships

Project Axis is designed to function not as an isolated local entity, but as a collaborative regional anchor that fosters strong inter-county partnerships across Southwest Georgia (SOWEGA).

- **The Cuthbert-Randolph County Axis (Probiotic Bone Meal Pipeline):** The city of Cuthbert and surrounding Randolph County represent a vital agricultural hub dominated by traditional crop production and regional poultry operations. Odynex Axis Logistics LLC will establish a dedicated distribution pipeline directly supplying Cuthbert's farmers and agricultural cooperatives with premium, localized probiotic bone meal.
- **Mutual Economic Benefits for Cuthbert:** By replacing expensive, synthetically manufactured fertilizers with our hyper-local, nutrient-dense organic bone meal, Cuthbert-area agricultural businesses can drastically reduce input costs. Furthermore, the inclusion of live beneficial soil microbes (inoculated during the Odynex Institute’s natural fermentation process) will enhance soil health and crop yields specifically for Randolph County's row crops.
- **Sourcing Partnership Network:** In exchange for finished fertilizer distribution, the SOWEGA Asian Pacific Chamber of Commerce and Foundation will coordinate cooperative agreements with agricultural processors in neighboring counties (including Randolph, Quitman, and Early counties). The logistics LLC will systematically retrieve underutilized agricultural byproducts—such as peanut hulls, crop trimmings, and processing waste—from these surrounding areas. This creates a mutually beneficial regional network: neighboring counties shed waste disposal liabilities while Fort Gaines gains free raw manufacturing inputs.

VI. Small-Town Self-Sustainment & Social Uplift

At its core, Project Axis is a blueprint for rural resilience. It proves that small towns like Fort Gaines do not need to import expensive synthetic resources or rely on outside industrial giants; rather, they can achieve economic independence by upcycling the abundant agricultural assets already present in Southwest Georgia.

- **Uplifting Communities with Better-Paying Opportunities:** Rural economies are frequently limited to low-wage retail or seasonal labor. Project Axis introduces specialized, high-utility technical careers directly into Clay County. By converting basic farming into a biotechnology operation, Odynex Axis Logistics LLC will

provide stable, year-round manufacturing, processing, and logistics jobs that pay significantly higher than the local baseline, keeping wealth localized within the community.

- **Youth Education & Workforce Empowerment:** To ensure long-term community transformation, the Odynex Institute—in tandem with the SOWEGA Asian Pacific Chamber Foundation—will launch dedicated youth educational initiatives. The facility will partner with local schools and regional technical colleges to offer hands-on STEM internships, agricultural technology workshops, and vocational training programs. Local youth will learn advanced biotechnology skills—such as microbial fermentation, automated vertical farming, and closed-loop mechanics—cultivating a highly skilled generation capable of leading the future of rural green manufacturing without needing to leave their hometowns for high-paying careers.

VII. Regional Economic Impact for Fort Gaines

- **Agricultural Waste Upcycling:** Turning local processing liabilities into industrial assets by sourcing thousands of tons of underutilized regional agricultural waste (peanut hulls, pecan shells, and crop trimmings).
- **Strategic Logistics Advantage:** Leveraging Fort Gaines' border location to frictionlessly ship high-value, non-live final after-products directly into the massive Georgia and Alabama agricultural pipelines.